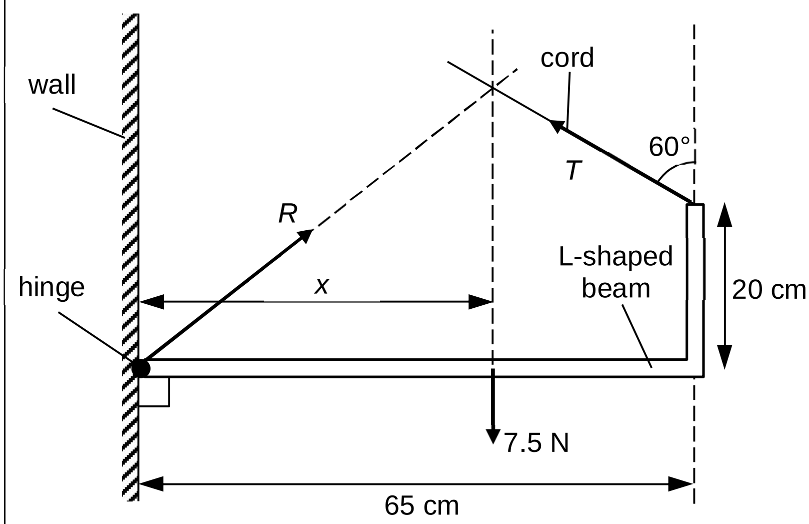
No.	Solution	Remarks
1(a)(i)	Since $T = kx$, using (0, 5) and (5.0, 25) $k = \dfrac{5.0 - 0}{(25-5)\times 10^{-2}} = 25\text{ N m}^{-1}$	[1] for substitution [1] for answer
(a)(ii)	$\text{elastic potential energy} = \dfrac{1}{2}kx^2$ $= \dfrac{1}{2}(25)(0.20)^2$ $= 0.50\text{ J}$	[1] for answer
(b)(i)		[1]
(b)(ii)	From Fig. 1.1, T = 5.0 N when extension = 20 cm. Resolving T vertically and horizontally, both components will lead to an anticlockwise moment about the hinge. Taking moments about the hinge, $(7.5)x = (5.0\sin 60°)(20) + (5.0\cos 60°)(65)$ $x = 33\text{ cm}$	[1] for correct T [1] for substitution [1] for answer
(b)(iii)	The system is in equilibrium, i.e. $F_{net} = 0$. Horizontally, $R_x = T_x = 5.0\sin 60° = 4.3\text{ N}$ Vertically, $R_y + T_y = 7.50\text{ N}$ $R_y = 7.5 - 5.0\cos 60°$ $= 5.0\text{ N}$ Hence, $R = \sqrt{4.3^2 + 5.0^2} = 6.6\text{ N}$	[1] for correct components of R [1] for correct R

2(a)(i)		[1] correct